

PostgreSQL Cascading Replication — Hands-on Lab

Kubernetes · Zalando Operator · Spilo · Patroni · PostgreSQL 18 · namespace postgres-ha · cluster acid-postgres-ha

Goal: change Replica-2 so it receives WAL from Replica-1 instead of the primary. Patroni tag replicatefrom is member-local. Do not put it under spec.patroni in the Zalando CR. A kubectl delete/recreate of that pod loses this edit.

Target topology

Always rediscover the primary. Examples below assume 0=primary, 1=upstream replica, 2=cascade replica.

```
Before: PRIMARY ---> REPLICA-1
        +--> REPLICA-2
```

```
After:  PRIMARY ---> REPLICA-1 ---> REPLICA-2
```

Lab 1 — Cluster present

```
kubectl get ns postgres-ha
kubectl get postgresql -n postgres-ha
kubectl get pods -n postgres-ha -l application=spilo
```

Lab 2 — Roles

```
kubectl get pods -n postgres-ha -l application=spilo -L spilo-role
```

Substitute real names if master is not acid-postgres-ha-0.

Lab 3 — Patroni list

```
kubectl exec -n postgres-ha acid-postgres-ha-0 -- patronictl list
```

Expect Leader + two Replicas in streaming.

Lab 4 — Both replicas follow the primary (before)

```
kubectl exec -n postgres-ha acid-postgres-ha-0 -- psql -U postgres -c \
"SELECT application_name, client_addr, state, sync_state, sent_lsn, replay_lsn FROM pg_stat_replication;"
```

Expect two rows (Replica-1 and Replica-2) streaming from the primary.

Lab 5–6 — WAL receiver on each replica

```
kubectl exec -n postgres-ha acid-postgres-ha-1 -- psql -U postgres -c \
"SELECT status, sender_host, sender_port, slot_name FROM pg_stat_wal_receiver;"
```

```
kubectl exec -n postgres-ha acid-postgres-ha-2 -- psql -U postgres -c \
"SELECT status, sender_host, sender_port, slot_name FROM pg_stat_wal_receiver;"
```

Record Replica-2 sender_host. After cascade it must change to Replica-1.

Lab 7–8 — Replica-1 can send WAL

```
kubectl exec -n postgres-ha acid-postgres-ha-1 -- psql -U postgres -c "SHOW hot_standby;"
kubectl exec -n postgres-ha acid-postgres-ha-1 -- psql -U postgres -c "SHOW max_wal_senders;"
kubectl exec -n postgres-ha acid-postgres-ha-1 -- bash -c "grep -i replication \$(psql -U postgres -Atc 'SHOW hba_file;')"
```

Lab 9 — REST API before change

```
kubectl exec -n postgres-ha acid-postgres-ha-2 -- curl -s http://localhost:8008/patroni
```

tags should not yet contain replicatefrom.

Lab 10–14 — Locate, backup, inspect Replica-2 Patroni YAML

```
kubectl exec -it -n postgres-ha acid-postgres-ha-2 -- bash
ps -ef | grep '[p]atroni'
find /run /etc -maxdepth 4 -type f \( -name '*patroni*.yaml' -o -name '*patroni*.yaml' \) 2>/dev/null
```

Do not assume the path. Typical Spilo path is /run/postgresql/patroni.yaml.

```
PATRONI_CONFIG=/run/postgresql/patroni.yaml
ls -l $PATRONI_CONFIG
cp $PATRONI_CONFIG ${PATRONI_CONFIG}.before-cascade
python3 -c "import yaml; print('PyYAML available')"
```

```
python3 -c "import yaml; c=yaml.safe_load(open('$PATRONI_CONFIG')); print(c.get('tags',{}))"
```

Lab 15–17 — Set replicatefrom and validate

```
python3 -c "import yaml; p='$PATRONI_CONFIG'; c=yaml.safe_load(open(p)); c.setdefault('tags',{}]['replicatefrom']='acid-postgres-ha-1'"
python3 -c "import yaml; c=yaml.safe_load(open('$PATRONI_CONFIG')); print(c.get('tags',{}))"
grep -A10 "^tags:" $PATRONI_CONFIG
patroni --validate-config $PATRONI_CONFIG --ignore-listen-port
```

Expected tag: replicatefrom: acid-postgres-ha-1. Stop if YAML validation fails.

Lab 18–20 — Reload and confirm tag

```
patronictl -c $PATRONI_CONFIG reload acid-postgres-ha acid-postgres-ha-2 --force
exit

kubect exec -n postgres-ha acid-postgres-ha-2 -- curl -s http://localhost:8008/patroni | grep replicatefrom
kubect logs -n postgres-ha acid-postgres-ha-2 --tail=100
```

Labs 22–24 — Three proofs the cascade is live

```
# Proof 1 — Replica-2 receives from Replica-1
kubect exec -n postgres-ha acid-postgres-ha-2 -- psql -U postgres -c \
"SELECT status, sender_host, sender_port FROM pg_stat_wal_receiver;"

# Proof 2 — Replica-1 sends to Replica-2
kubect exec -n postgres-ha acid-postgres-ha-1 -- psql -U postgres -c \
"SELECT application_name, client_addr, state FROM pg_stat_replication;"

# Proof 3 — Primary no longer lists Replica-2 as a direct consumer
kubect exec -n postgres-ha acid-postgres-ha-0 -- psql -U postgres -c \
"SELECT application_name, client_addr, state FROM pg_stat_replication;"
```

Lab is not complete until all three proofs match Primary → Replica-1 → Replica-2.

Labs 25–34 — Data path test

```
kubect exec -n postgres-ha acid-postgres-ha-0 -- psql -U postgres -c "CREATE DATABASE cascade_lab;"
kubect exec -n postgres-ha acid-postgres-ha-0 -- psql -U postgres -d cascade_lab -c "
CREATE TABLE cascade_test (
  id BIGSERIAL PRIMARY KEY,
  message VARCHAR(200),
  created_at TIMESTAMPTZ DEFAULT CURRENT_TIMESTAMP
);
INSERT INTO cascade_test(message) VALUES ('First cascading replication transaction');
INSERT INTO cascade_test(message) SELECT 'Cascade transaction ' || generate_series(1,10000);
SELECT COUNT(*) FROM cascade_test;
"

kubect exec -n postgres-ha acid-postgres-ha-1 -- psql -U postgres -d cascade_lab -c "SELECT COUNT(*) FROM cascade_test;"
kubect exec -n postgres-ha acid-postgres-ha-2 -- psql -U postgres -d cascade_lab -c "SELECT COUNT(*) FROM cascade_test;"
```

After catch-up all three counts match. Replica-2 may trail Replica-1 by a small LSN gap.

```
kubect exec -n postgres-ha acid-postgres-ha-1 -- psql -U postgres -c "SELECT pg_last_wal_receive_lsn(), pg_last_wal_replay_lsn();"
kubect exec -n postgres-ha acid-postgres-ha-2 -- psql -U postgres -c "SELECT pg_last_wal_receive_lsn(), pg_last_wal_replay_lsn();"
```

Labs 35–39 — Upstream blip (optional)

Do not delete Replica-1 or Replica-2 pods: the cascade edit on Replica-2 is only in that container's Patroni file. Watch logs and sender_host if the instructor simulates an upstream outage. Patroni may fall back to the primary if the tagged source is unusable, then prefer Replica-1 again when it returns.

Labs 41–43 — Why cascade vs sync

Cascade answers: where does this standby get WAL? Sync answers: when may COMMIT return? You can combine: sync to Replica-1, cascade async from Replica-1 to Replica-2. WAN pattern: one stream across the WAN, then LAN fan-out.

Labs 44–46 — Roll back

```
kubect exec -it -n postgres-ha acid-postgres-ha-2 -- bash
PATRONI_CONFIG=/run/postgresql/patroni.yml
cp ${PATRONI_CONFIG}.before-cascade $PATRONI_CONFIG
patronictl -c $PATRONI_CONFIG reload acid-postgres-ha acid-postgres-ha-2 --force
exit
```

```
kubectl exec -n postgres-ha acid-postgres-ha-2 -- psql -U postgres -c \
"SELECT status, sender_host FROM pg_stat_wal_receiver;"
kubectl exec -n postgres-ha acid-postgres-ha-0 -- patronictl list
```

Replica-2 sender_host should return to the primary. Topology is again Primary → both replicas.

Troubleshooting

Problem	Check
replicatefrom missing after edit	grep tags in YAML, reload with -c \$PATRONI_CONFIG, curl :8008/patroni
Replica-2 still follows primary	Replica-1 must be in recovery and running; wait and re-check sender_host
Replica-1 cannot send WAL	hot_standby=on, max_wal_senders>0, replication lines in pg_hba
auth failure in Replica-1 logs	no pg_hba.conf entry / password; inspect hba_file grep replication
Pod recreated, cascade gone	Expected. Re-apply Labs 10–18. This is not CR-persistent.

Pass criteria

1) Replica-2 pg_stat_wal_receiver.sender_host is Replica-1. 2) Replica-1 pg_stat_replication lists Replica-2. 3) Primary pg_stat_replication does not list Replica-2 while cascade is on. 4) cascade_lab.cascade_test counts match after catch-up. 5) Rollback restored direct streaming.